# Deep Learning Model Used- DistilBERT (Distilled Bidirectional Encoder Representations from Transformers)

DistilBERT is a smaller, faster, and lighter version of the well-known BERT (Bidirectional Encoder Representations from Transformers) model. It was developed using a technique called knowledge distillation, where a large model (BERT) transfers its knowledge to a smaller one (DistilBERT).

The key idea is to retain most of BERT's powerful language understanding ability while reducing the size and computation requirements. This makes DistilBERT a great choice for real-time applications like automated review rating, where speed and efficiency are as important as accuracy.

## 1.Why DistilBERT Model

- **Efficiency**: DistilBERT is a lighter, faster version of BERT. It retains 97% of BERT's performance while being 40% smaller and 60% faster, making it ideal for real-time applications like review rating.
- **Transfer Learning**: Pretrained on a large corpus, it already understands language patterns, reducing the need for massive datasets.
- **Generalization**: It handles unseen data well, as it has strong semantic understanding.
- **Resource-Friendly**: Suitable for GPU/CPU with limited memory, unlike heavier models (BERT-large, GPT-based).

## 2.Model Architecture Overview

- **Base Architecture**: DistilBERT is based on the Transformer Encoder architecture.
- **Key Components**:
  - **Token Embeddings**: Converts input text into vector representations.
  - **Positional Embeddings**: Adds order information since Transformers lack sequential awareness.
  - **Multi-Head Self-Attention Layers**: Learn contextual relationships between words.
  - **Feed-Forward Layers**: Transform attention outputs into deeper representations.
  - **Classification Head**: A linear layer on top of DistilBERT's pooled output, producing logits for star ratings.

- **Layers**: 6 transformer encoder layers (vs. 12 in BERT-base).
- **Hidden Size**: 768 dimensions.
- **Attention Heads**: 12 attention heads.
- **Parameters**: ~66M parameters (compared to ~110M in BERT-base).

## 3.Workflow of Model

1. Input Text
    - Raw review text is tokenized using DistilBERT tokenizer (WordPiece subwords).
    - Generates input_ids and attention_mask.
2. Embedding Layer
    - Tokens are mapped to dense vectors with positional encoding.
3. Transformer Encoder Layers
    - Multiple layers of self-attention capture context (e.g., "bad service" → negative sentiment).
4. Classification Head
    - The [CLS] token representation is passed through a dense layer to predict one of 5 classes (ratings 1–5).
5. Output
    - Model outputs logits → converted to predicted star rating.

# 4.Training Details

- **Dataset**: Preprocessed reviews labeled with star ratings.
- **Train/Test Split**: 80/20 split with stratified sampling.
- **Tokenizer**: DistilBertTokenizer with max length = 64.
- **Model**: DistilBertForSequenceClassification with num_labels=5.
- **Optimizer**: AdamW (learning rate = 2e-5).
- **Batch Size**: 16.
- **Epochs**: 10 (with **early stopping** to avoid overfitting).
- **Device**: Trained on GPU (CUDA).
- **Best Model Saving**: Model checkpoint saved when validation loss improved.

## 5.Advantages

- Lightweight yet powerful → faster inference, suitable for real-time applications.
- Pretrained knowledge → strong contextual understanding, even with limited training data.
- Handles long dependencies better than RNN/LSTM.
- General-purpose → effective across multiple NLP tasks (classification, sentiment, Q&A).

## 6.Limitations

- **Data Hungry**: Needs large, diverse data for best fine-tuning performance.
- **Limited Sequence Length**: Can only handle up to 512 tokens (we limited to 64).
- **Ambiguity in Mid-Ratings**: Struggles to differentiate middle ratings (2, 3, 4) due to overlapping language.
- **Interpretability**: Transformer decisions are hard to explain compared to simpler models.